

Writing Reports, Proposals, and Technical Documents

By Laurie A. Pinkert



Presentation Overview

- Genres and constraints
- Scientific style
 - Document design
 - Textual features
- Writing and revision
 - Principles
 - An exercise

Genres and Constraints

- Reports
- Proposals
- Other technical documents (abstracts, etc.)

What is Style?

- Document design
 - Formatting standards
- Rules or guidelines
 - Writing principles

Disciplinary Differences

Best Practices in the Humanities:

- Flexible conventions for structure depending on the task; transitions, rather than headings, are commonly used to demarcate major sections of writing
- Reliance of text rather than visual elements to convey main ideas
- Transitions generally signify changes in or two emphasize particular aspects of the author's position ("therefore," "furthermore," in any case")
- Use MLA or other appropriate humanities styles to cite sources

Best Practices in the Sciences:

- Strict adherence to structural genre conventions, such as the abstract, introduction, methods, results, references
- Use multiple visual elements, such as charts or graphs, to demonstrate important concepts
- Clear procedural transitions that signal particular steps in an experiment (e.g., first, second, third)
- Use of a discipline-appropriate scientific style of citation

Document Design

- Headers
 - Use distinguishable features (e.g., bold, all caps, larger font, etc.)
- Spacing between sections
 - Should be consistent
- Each textual unit should use the same font and size
 - Headers should all be the same font and size
 - Body text should all be the same font and size
 - These sections may differ from one another

Document Design

- Tables and charts should be
 - All the same size
 - Labeled consistently
 - Scaled appropriately
 - Embedded in text or located in an appendix
 - Placed immediately after the paragraph that first refers to it
 - Referred to in text

Simple Table

Example of a simple table

Table 1. Effect of ciliary neurotrophic factor (CNTF) on protein content and lysosomal latency in proximally denervated soleus.

Injection	Protein (mg/muscle)	Lysosomal latency (%)
Vehicle	8.8 ± 0.9	43 ± 4
CNTF	8.0 ± 0.8*	33 ± 3*

Proximally denervated muscles were injected (4 µl/muscle/100g body weight) daily, beginning immediately after denervation, with vehicle only (PBS with 0.1% bovine serum albumin) or CNTF (5 µg/ml PBS). At 3 days after denervation, protein content (Biuret method) and lysosomal latency were determined as described in Materials and Methods.

*Significantly different from vehicle only (P<0.02).

16

* Sample from p. 16 <<http://www.biochem.arizona.edu/marc/Sci-Writing.pdf>>

Complex Table

18 / 25 150% Find

Example of a complex table

Table 2. Final muscle weights, total protein concentration, GLUT-4 protein, and hexokinase and citrate synthase activities in skeletal muscle following 3-day or 7-day hindlimb suspension (HS).

Group	Final Muscle Weight (g/100 g BW)	Total Protein Concentration (mg/g)	Total GLUT-4 (% of control)	Total Hexokinase (nmol/mg/min)	Citrate Synthase
Soleus					
Weight-bearing	40.0 ± 0.8	149 ± 3	100 ± 9	1.5 ± 0.2	107 ± 5
3-day HS	33.6 ± 1.3*	133 ± 2*	132 ± 16	2.1 ± 0.1*	122 ± 6
7-day H	21.8 ± 1.0*	130 ± 7*	162 ± 10*	2.9 ± 0.2*	143 ± 6*
EDL					
Weight-bearing	42.7 ± 0.7	158 ± 3	100 ± 7	1.9 ± 0.1	64 ± 3
3-day HS	45.9 ± 0.8	156 ± 5	87 ± 7	2.0 ± 0.2	69 ± 7
7-day H	42.4 ± 2.4	146 ± 8	112 ± 8	2.9 ± 0.1*	57 ± 7

Values are means ± SE for 8-17 animals in the pooled weight-bearing group and 6-18 animals in the respective HS groups. * P<0.05 vs. weight-bearing control group.

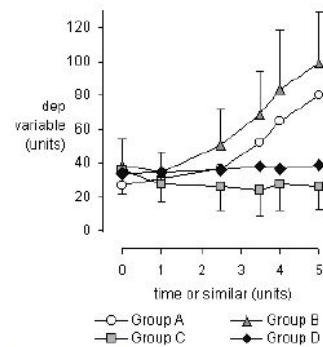
8.50 x 11.00 in

* Sample from p. 18 <<http://www.biochem.arizona.edu/marc/Sci-Writing.pdf>>

Figures

See the examples (Figures 1-4).

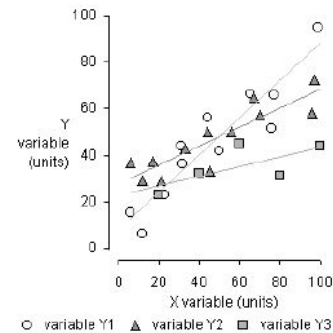
Figure 1: Informative title for a time series^a.



Data are means. Bars are standard deviations (shown only for Groups B and C).

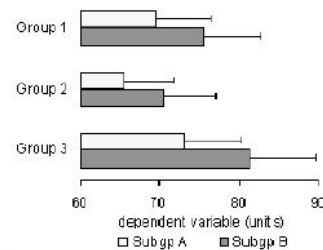
^aUse letters to label footnotes, if necessary.

Figure 2: Informative title for a scattergram



Least-squares lines are shown for each variable.

Figure 3: Informative title for a bar graph.



Data are means. Bars are standard deviations.

* Sample from p. 19 <<http://www.biochem.arizona.edu/marc/Sci-Writing.pdf>>

Principles for Writing Clearly

1. Open your sentences with short concrete subjects that name the characters.
2. Use specific verbs to name their important actions.
3. Get to the main verb quickly.
4. Open your sentences with information familiar to reader.
5. Push new, complex units of information to the end of the sentence.
6. Begin sentences constituting a passage with consistent topic/subjects.
7. Be concise.
8. Control sprawl.
9. Use parallel structures.
10. Above all, write to others as you have others write to you.

* Principles from Joseph Williams's (2012) *Style: The Basics of Clarity and Grace*

#4 - Start with Familiar Information

By starting with the familiar, you should be able to make connections between your findings and your conclusions

- **TRY:** After identifying the lethal allele, we modified our selections in the next round.
- **Not:** The lethal allele changed my strategy.

Language to Create Connections

- Addition

“Furthermore,” “In addition,” “Moreover”

- Elaboration

“By extension,” “In short,” “In other words”

- Example

“For instance,” “For example,” “As an illustration”

- Comparison

“Likewise,” “Similarly”

- Cause and effect

“Accordingly,” “Consequently,” “Hence,” “As a result”

#7 – Be Concise but Clear

- Use plain language

Scientific reports are fairly straightforward, so avoid metaphors, similes, allusions, etc.

- **TRY:** The selections in round two were informed by my prior experience.
- **NOT:** Making selections was easy as pie.

#6 - Begin with Consistent Subjects

Topics are crucial for readers because readers depend on topics to focus their attention on particular ideas toward the beginning of sentences. Topics tell readers what a whole passage is "about." If readers feel that a sequence of topics is coherent, then they will feel they are moving through a paragraph from a cumulatively coherent point of view. But if throughout the paragraph readers feel that its topics shift randomly, then they have to begin each sentence out of context, from no coherent point of view. When that happens, readers feel dislocated, disoriented, and out of focus.

#2 – Use Specific Verbs

Topics are crucial for readers because readers depend on topics to focus their attention on particular ideas toward the beginning of sentences. Topics tell readers what a whole passage is "about." If readers feel that a sequence of topics is coherent, then they will feel they are moving through a paragraph from a cumulatively coherent point of view. But if throughout the paragraph readers feel that its topics shift randomly, then they have to begin each sentence out of context, from no coherent point of view. When that happens, readers feel dislocated, disoriented, and out of focus.

Additional Resources

- Visit the Purdue Online Writing Lab:
<http://owl.english.purdue.edu>
- Email brief questions to the OWL Mail Tutors:
<http://owl.english.purdue.edu/contact/owlmailtutors>
- Publication Manual of the American Psychological Association
- Scientific Writing Guide:
<https://cbc.arizona.edu/sites/cbc.arizona.edu/files/marc/Sci-Writing.pdf>

The End

